task 9

Primary ALB in Region A

- Secondary ALB in Region B
- Health checks
- Automatic failover

STEP 1: Setup regions

Go to Region A - singapore ap-southeast-1

Go to Region B - sydney ap-southeast-2

1. Launch EC2 instances

Use Amazon EC2

- Launch 2 instances
- Allow HTTP (port 80) in Security Group

2. Install web server

SSH into each:

step 3

Create:

- 1 Application Load Balancer in **Singapore (Primary)**
- 1 ALB in **Sydney (Secondary)**

Each ALB → points to its EC2

step 3. Create Health Check (Route 53)

- Type: HTTP
- Endpoint: Primary ALB DNS

step 4 : Create Hosted Zone + connect domain

- Create hosted zone in Route 53
- Copy NS records
- Paste them in Hostinger nameservers

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step 5. Create Failover Records

. Create Hosted Zone + connect domain

- Create hosted zone in Route 53
- Copy NS records
- Paste them in Hostinger nameservers

---

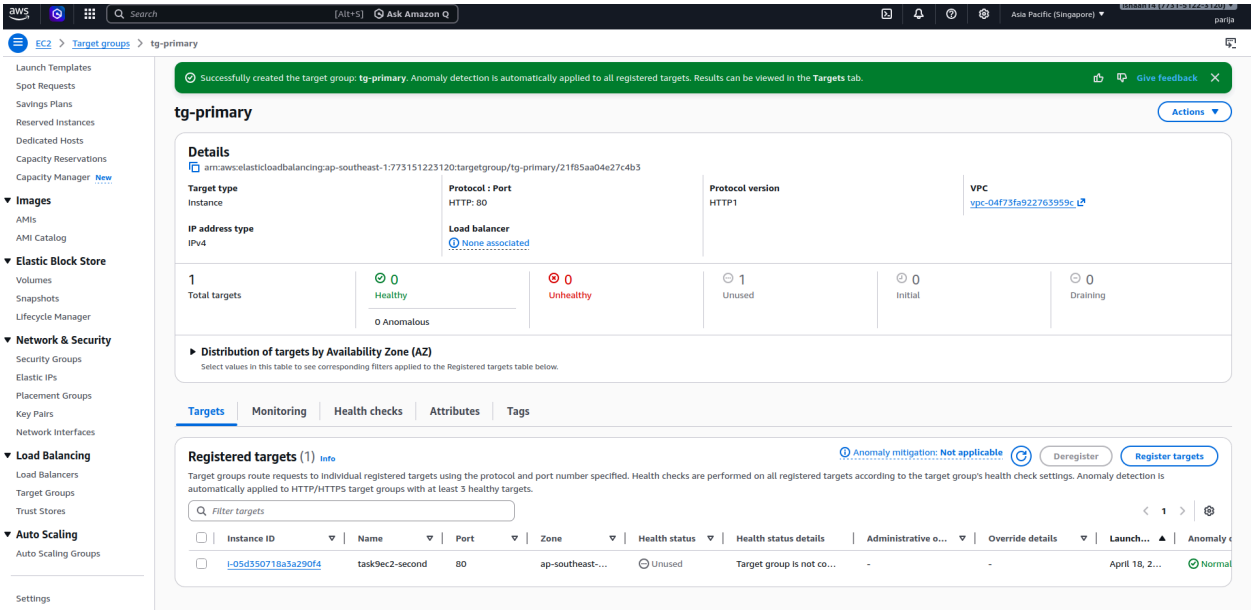
step 6 Create Failover Records

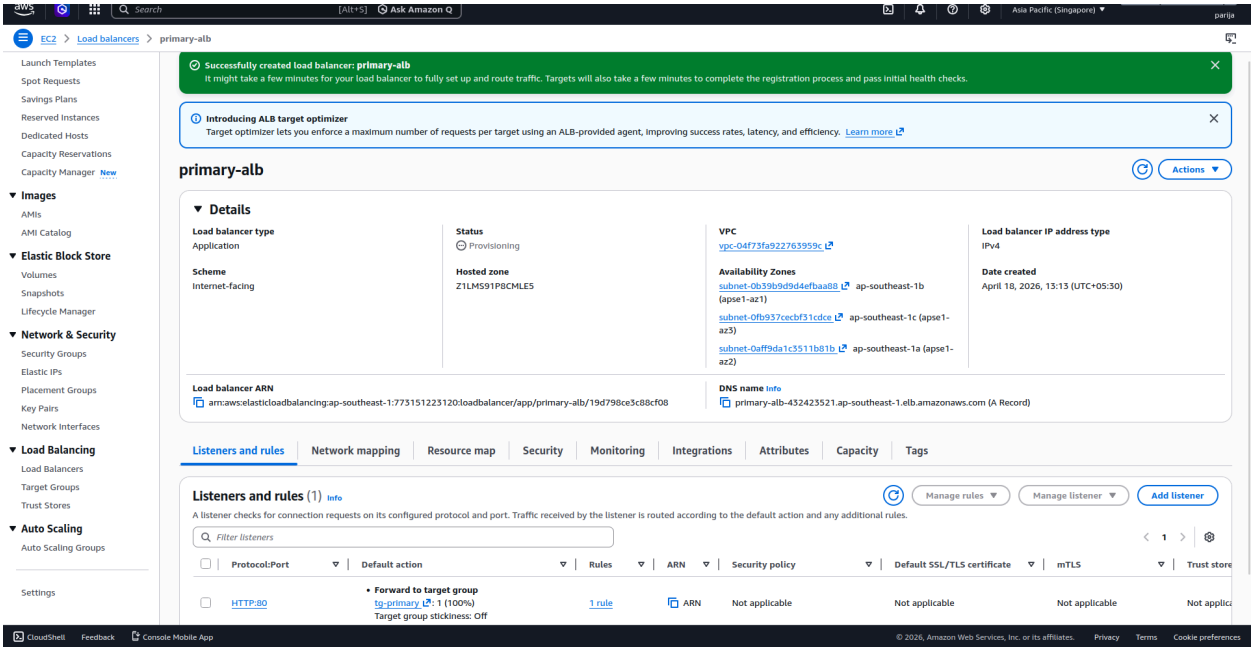
- Primary record:
  - Alias → Primary ALB
  - Routing → Failover (Primary)
  - Attach health check
- Secondary record:
  - Alias → Secondary ALB
  - Routing → Failover (Secondary)
- Primary record:

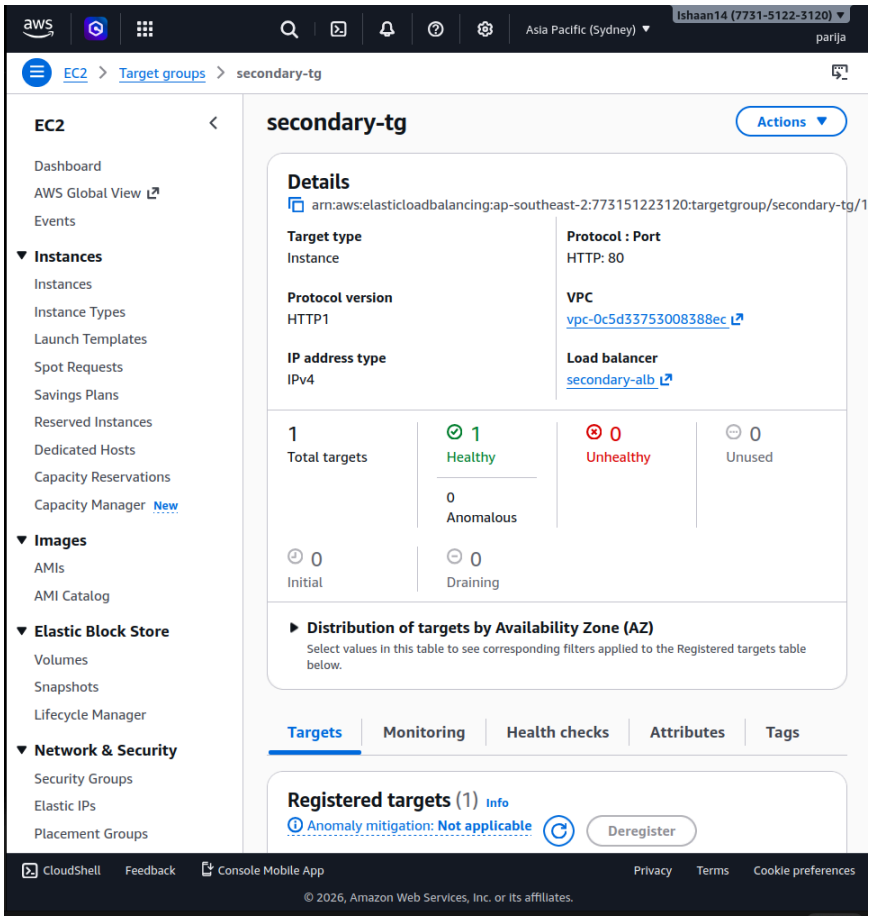
    - Alias → Primary ALB
    - Routing → Failover (Primary)
    - Attach health check
- Secondary record:
    - Alias → Secondary ALB
    - Routing → Failover (Secondary)

```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 48 not upgraded.
Need to get 565 kB of archives.
After this operation, 1596 kB of additional disk space will be used.
Get:1 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-commo
n all 1.24.0-2ubuntu7.6 [43.5 kB]
Get:2 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64
 1.24.0-2ubuntu7.6 [521 kB]
Fetched 565 kB in 0s (20.3 MB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 72566 files and directories currently installed.)
Preparing to unpack .../nginx-common_1.24.0-2ubuntu7.6_all.deb ...
Unpacking nginx-common (1.24.0-2ubuntu7.6) ...
Selecting previously unselected package nginx.
Preparing to unpack .../nginx_1.24.0-2ubuntu7.6_amd64.deb ...
Unpacking nginx (1.24.0-2ubuntu7.6) ...
Setting up nginx-common (1.24.0-2ubuntu7.6) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/s
ystem/nginx.service.
Setting up nginx (1.24.0-2ubuntu7.6) ...
 * Upgrading binary nginx                                                                [ OK ]
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-4-42:~$ echo "PRIMARY SERVER" | sudo tee /var/www/html/index.html
PRIMARY SERVER
ubuntu@ip-172-31-4-42:~$
```

```
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 48 not upgraded.
Need to get 565 kB of archives.
After this operation, 1596 kB of additional disk space will be used.
Get:1 http://ap-southeast-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-commo
n all 1.24.0-2ubuntu7.6 [43.5 kB]
Get:2 http://ap-southeast-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64
 1.24.0-2ubuntu7.6 [521 kB]
Fetched 565 kB in 0s (17.8 MB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 72566 files and directories currently installed.)
Preparing to unpack .../nginx-common_1.24.0-2ubuntu7.6_all.deb ...
Unpacking nginx-common (1.24.0-2ubuntu7.6) ...
Selecting previously unselected package nginx.
Preparing to unpack .../nginx_1.24.0-2ubuntu7.6_amd64.deb ...
Unpacking nginx (1.24.0-2ubuntu7.6) ...
Setting up nginx-common (1.24.0-2ubuntu7.6) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/s
ystem/nginx.service.
Setting up nginx (1.24.0-2ubuntu7.6) ...
 * Upgrading binary nginx                                                 [ OK ]
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-43-234:~$ echo "PRIMARY SERVER (SINGAPORE)" | sudo tee /var/www/html/index.ht
ml
PRIMARY SERVER (SINGAPORE)
ubuntu@ip-172-31-43-234:~$
```




EC2 > Load balancers > primary-alb
Successfully created load balancer: primary-alb
It might take a few minutes for your load balancer to fully set up and route traffic. Targets will also take a few minutes to complete the registration process and pass initial health checks.
Introducing ALB target optimizer
Target optimizer lets you enforce a maximum number of requests per target using an ALB-provided agent, improving success rates, latency, and efficiency. Learn more
primary-alb
Actions
Details
Load balancer type
Application
Status
Provisioning
VPC
vpc-04f73fa922763959c
Load balancer IP address type
IPv4
Scheme
Internet-facing
Hosted zone
Z1LMS91P8CMLE5
Availability Zones
subnet-0b39b9d9d4efbaa88 ap-southeast-1b (apse1-az1)
subnet-0fb937cecbf31cdce ap-southeast-1c (apse1-az3)
subnet-0aff9da1c3511b81b ap-southeast-1a (apse1-az2)
Date created
April 18, 2026, 13:13 (UTC+05:30)
Load balancer ARN
arn:aws:elasticloadbalancing:ap-southeast-1:773151223120:loadbalancer/app/primary-alb/19d798ce3c88cf08
DNS name Info
primary-alb-432423521.ap-southeast-1.elb.amazonaws.com (A Record)
Listeners and rules
Network mapping
Resource map
Security
Monitoring
Integrations
Attributes
Capacity
Tags
Listeners and rules (1) Info
Manage rules
Manage listener
Add listener
A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the default action and any additional rules.
Filter listeners
Protocol:Port
Default action
Rules
ARN
Security policy
Default SSL/TLS certificate
mTLS
Trust store
HTTP:80
Forward to target group
tg-primary: 1 (100%)
Target group stickiness: Off
1 rule
ARN
Not applicable
Not applicable
Not applicable
Not applica


Asia Pacific (Sydney)
Ishaan14 (7731-5122-3120)
parija
EC2 > Target groups > secondary-tg
secondary-tg
Actions
Details
arn:aws:elasticloadbalancing:ap-southeast-2:773151223120:targetgroup/secondary-tg/1
Target type
Instance
Protocol : Port
HTTP: 80
Protocol version
HTTP1
VPC
vpc-0c5d33753008388ec
IP address type
IPv4
Load balancer
secondary-alb
1
Total targets
1
Healthy
0
Anomalous
0
Unhealthy
0
Unused
0
Initial
0
Draining
Distribution of targets by Availability Zone (AZ)
Select values in this table to see corresponding filters applied to the Registered targets table below.
Targets
Monitoring
Health checks
Attributes
Tags
Registered targets (1) Info
Anomaly mitigation: Not applicable
Deregister
CloudShell
Feedback
Console Mobile App
Privacy
Terms
Cookie preferences
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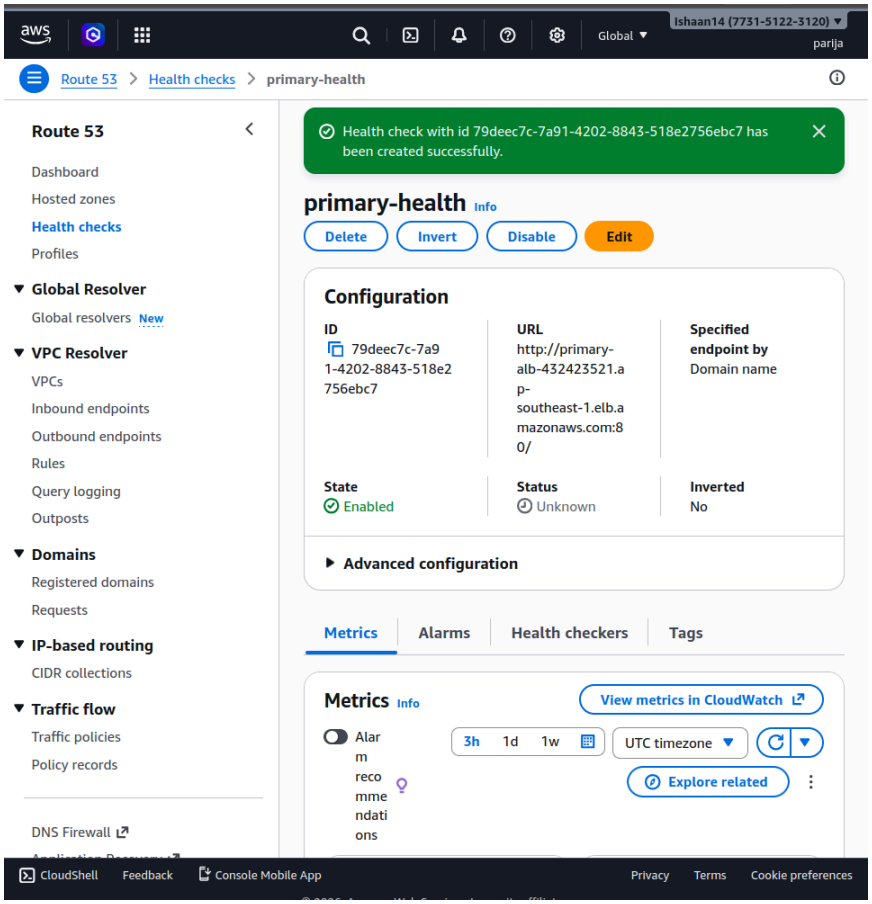
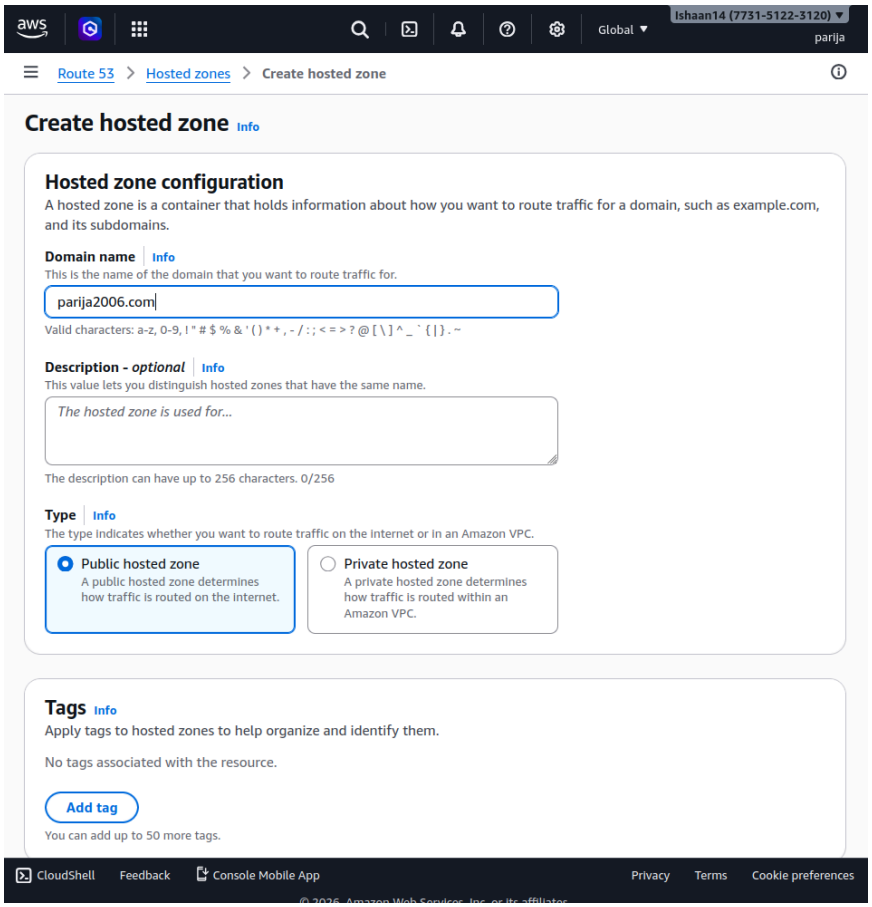
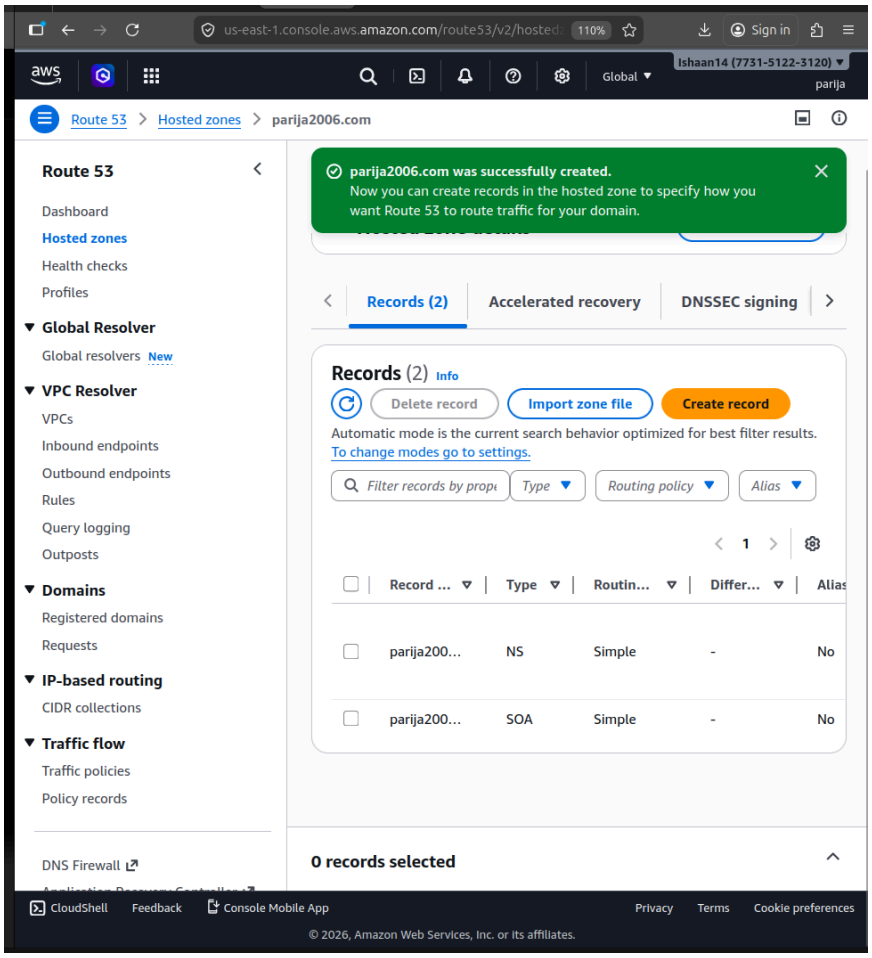
FINAL STEP: Route 53 (Failover)




aws
Global
Ishaan14 (7731-5122-3120)
parija
Route 53 > Hosted zones > Create hosted zone
Create hosted zone Info
Hosted zone configuration
A hosted zone is a container that holds information about how you want to route traffic for a domain, such as example.com, and its subdomains.
Domain name Info
This is the name of the domain that you want to route traffic for.
parija2006.com
Valid characters: a-z, 0-9, ! " # $ % & ' ( ) * + , - / : ; < = > ? @ [\ ] ^ _ ` { | } . ~
Description - optional Info
This value lets you distinguish hosted zones that have the same name.
The hosted zone is used for...
The description can have up to 256 characters. 0/256
Type Info
The type indicates whether you want to route traffic on the Internet or in an Amazon VPC.
Public hosted zone
A public hosted zone determines how traffic is routed on the Internet.
Private hosted zone
A private hosted zone determines how traffic is routed within an Amazon VPC.
Tags Info
Apply tags to hosted zones to help organize and identify them.
No tags associated with the resource.
Add tag
You can add up to 50 more tags.
CloudShell
Feedback
Console Mobile App
Privacy
Terms
Cookie preferences


us-east-1.console.aws.amazon.com/route53/v2/hosted
Sign in
Global
Ishaan14 (7731-5122-3120)
parija
Route 53 > Hosted zones > parija2006.com
Route 53
Dashboard
Hosted zones
Health checks
Profiles
Global Resolver
Global resolvers New
VPC Resolver
VPCs
Inbound endpoints
Outbound endpoints
Rules
Query logging
Outposts
Domains
Registered domains
Requests
IP-based routing
CIDR collections
Traffic flow
Traffic policies
Policy records
DNS Firewall
parija2006.com was successfully created.
Now you can create records in the hosted zone to specify how you want Route 53 to route traffic for your domain.
Records (2)
Accelerated recovery
DNSSEC signing
Records (2) Info
Delete record
Import zone file
Create record
Automatic mode is the current search behavior optimized for best filter results.
To change modes go to settings.
Filter records by prope
Type
Routing policy
Alias
Record ...
Type
Routin...
Differ...
Alias
parija200...
NS
Simple
-
No
parija200...
SOA
Simple
-
No
0 records selected
CloudShell
Feedback
Console Mobile App
Privacy
Terms
Cookie preferences
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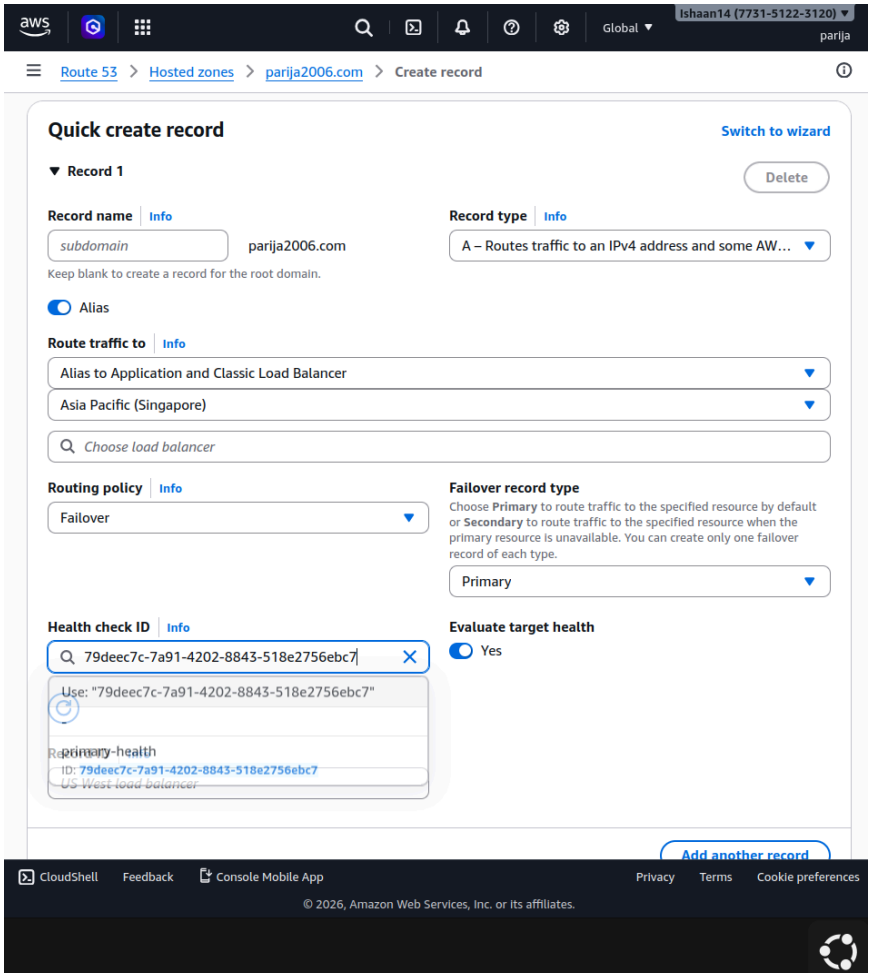
aws
Global
Ishaan14 (7731-5122-3120)
parija
Route 53 > Hosted zones > parija2006.com > Create record
Quick create record
Switch to wizard
Record 1
Delete
Record name Info
subdomain
parija2006.com
Keep blank to create a record for the root domain.
Record type Info
A – Routes traffic to an IPv4 address and some AW...
Alias
Route traffic to Info
Alias to Application and Classic Load Balancer
Asia Pacific (Singapore)
Choose load balancer
Routing policy Info
Failover
Failover record type
Choose Primary to route traffic to the specified resource by default or Secondary to route traffic to the specified resource when the primary resource is unavailable. You can create only one failover record of each type.
Primary
Health check ID Info
79deec7c-7a91-4202-8843-518e2756ebc7
Use: "79deec7c-7a91-4202-8843-518e2756ebc7"
primary-health
ID: 79deec7c-7a91-4202-8843-518e2756ebc7
Evaluate target health
Yes
Add another record
CloudShell
Feedback
Console Mobile App
Privacy
Terms
Cookie preferences
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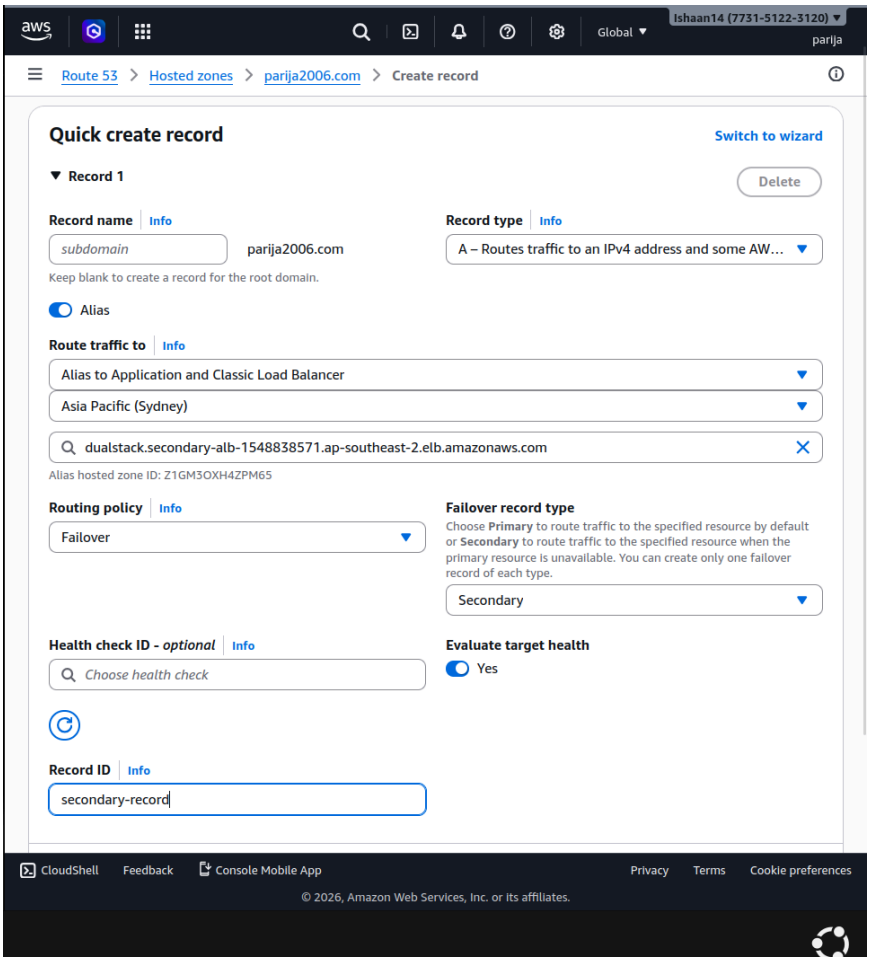
aws
Global
Ishaan14 (7731-5122-3120)
parija
Route 53 > Hosted zones > parija2006.com > Create record
Quick create record
Switch to wizard
Record 1
Delete
Record name Info
subdomain
parija2006.com
Keep blank to create a record for the root domain.
Record type Info
A – Routes traffic to an IPv4 address and some AW...
Alias
Route traffic to Info
Alias to Application and Classic Load Balancer
Asia Pacific (Sydney)
dualstack.secondary-alb-1548838571.ap-southeast-2.elb.amazonaws.com
Alias hosted zone ID: Z1GM3OXH4ZPM65
Routing policy Info
Failover
Failover record type
Choose Primary to route traffic to the specified resource by default or Secondary to route traffic to the specified resource when the primary resource is unavailable. You can create only one failover record of each type.
Secondary
Health check ID - optional Info
Choose health check
Evaluate target health
Yes
Record ID Info
secondary-record
CloudShell
Feedback
Console Mobile App
Privacy
Terms
Cookie preferences
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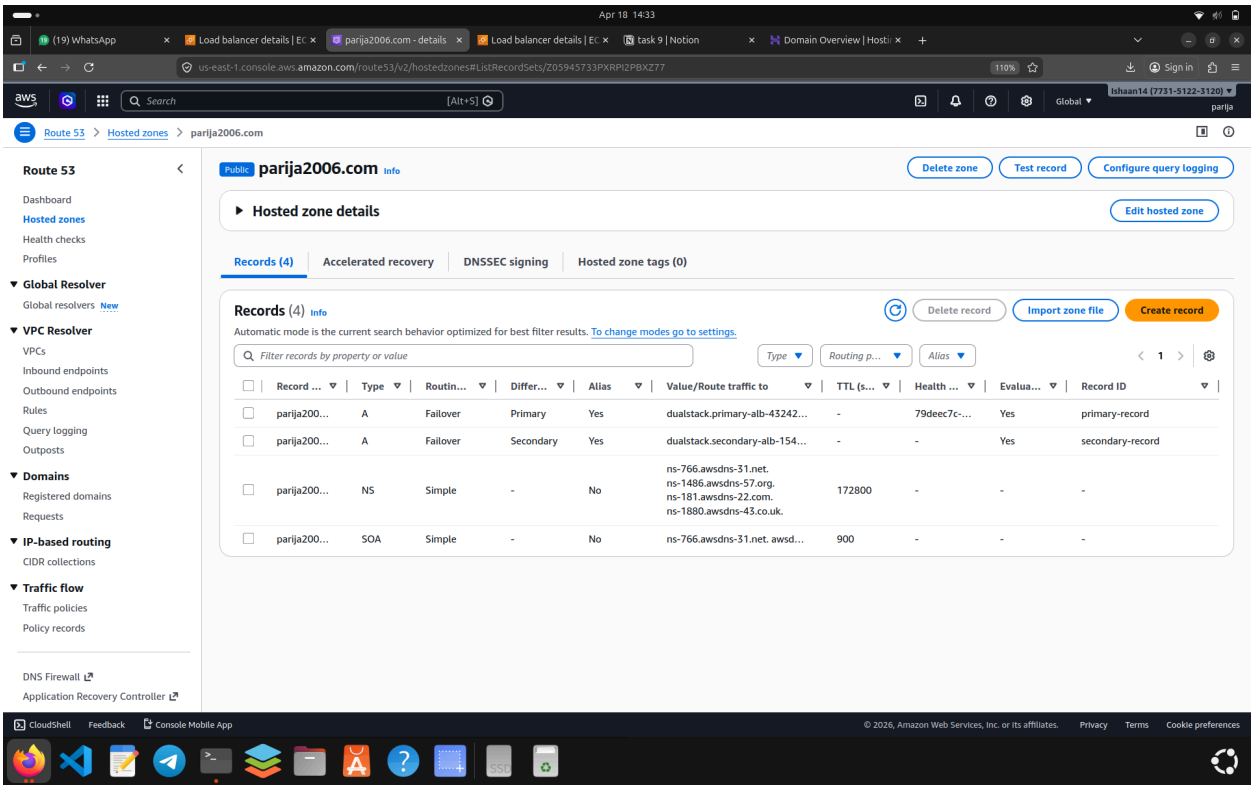
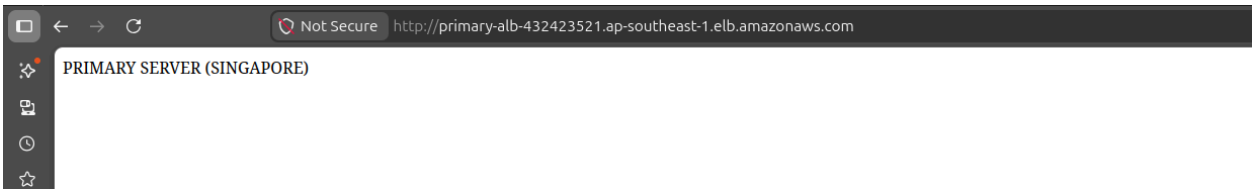
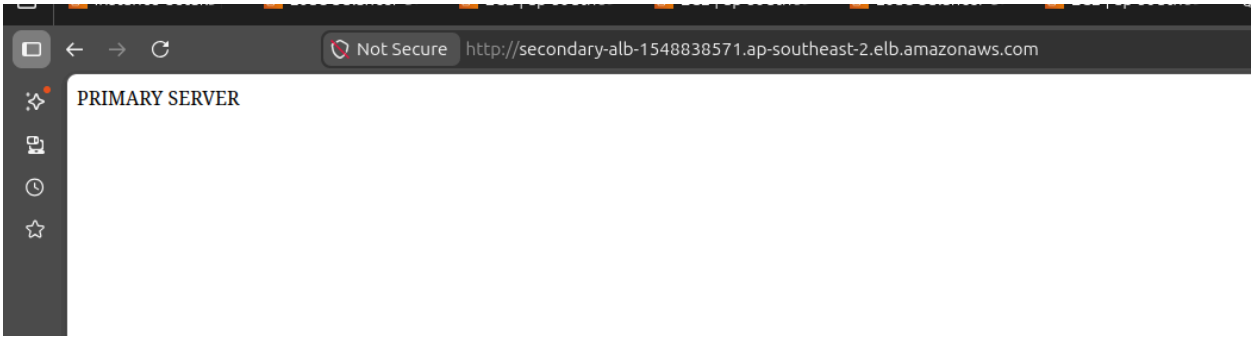

Route 53 > Hosted zones > parija2006.com
Public parija2006.com Info
Delete zone
Test record
Configure query logging
Hosted zone details
Edit hosted zone
Records (4)
Accelerated recovery
DNSSEC signing
Hosted zone tags (0)
Records (4) Info
Delete record
Import zone file
Create record
Automatic mode is the current search behavior optimized for best filter results. To change modes go to settings.
Filter records by property or value
Record ... | Type | Routin... | Differ... | Alias | Value/Route traffic to | TTL (s... | Health ... | Evalua... | Record ID
parija200... | A | Failover | Primary | Yes | dualstack.primary-alb-43242... | - | 79deec7c-... | Yes | primary-record
parija200... | A | Failover | Secondary | Yes | dualstack.secondary-alb-154... | - | - | Yes | secondary-record
parija200... | NS | Simple | - | No | ns-766.awsdns-31.net. ns-1486.awsdns-57.org. ns-181.awsdns-22.com. ns-1880.awsdns-43.co.uk. | 172800 | - | - | -
parija200... | SOA | Simple | - | No | ns-766.awsdns-31.net. awsd... | 900 | - | - | -


Not Secure
http://primary-alb-432423521.ap-southeast-1.elb.amazonaws.com
PRIMARY SERVER (SINGAPORE)


Not Secure
http://secondary-alb-1548838571.ap-southeast-2.elb.amazonaws.com
PRIMARY SERVER